LFE-SCCM: Low-Frequency Enhanced Method for Screen Content Image Compression

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Abstract. In the context of remote working and studying, the demand for efficient compression of screen content images has significantly increased. However, current image compression models primarily focus on natural images, neglecting the increasing need for compressing screen content images. To address this issue, we propose a new learning-based compression model of screen content images inspired by existing learning-based natural image compression models and combined with characteristics of screen content images. Our approach fully considers low-frequency information, which takes up a large portion of everyday screen use cases, although many researchers highlighted that high-frequency information mainly affects the perceptual quality of screen content images and videos. We integrate a Frequency Separator Module (FSM) and a specialized low-frequency enhancement branch into our model. The proposed low-frequency enhanced method, as shown by experimental results, outperforms existing hyperprior-based models in compressing screen content. The source code for our model is publicly accessible at https://github.com/RoneChen/LFE-SCCM.

Keywords: Image Compression · Image Processing · Screen Content Image Compression.

1 Introduction

In recent years, the rise of remote work and social networks has led to a significant increase in the proportion of images and videos used in daily communication. As deep learning continues to thrive in computer vision and related domains, there is growing interest among researchers in employing Deep Neural Networks (DNNs) to supplant traditional techniques such as JPEG, WebP, and VVC in image compression. This technology, which is foundational and widely used in image processing, is increasingly attracting the attention of the research community. However, most of them overlook the growing data volume of screen content images, especially in remote working and social media, screenshots are widely used to transfer and clarify information. In this case, an efficient screen content

coding method is highly desirable. Moreover, according to the characteristics of screen contents, generated sharp edges (high frequency) and large areas with the same color (low frequency), using the same strategy to deal with both of these information might have a great potential to be improved. LC-FDNet, a learned-based image compression method that uses low-frequency information as a strong prior for high-frequency compression, presents a good performance on natural images [29]. Inspired by that design and the analysis of conventional compression algorithms, we believe that screen content images will get more advantages from this scheme.

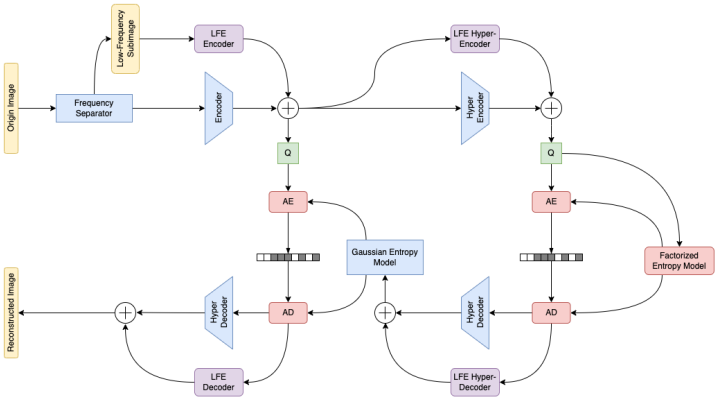


Fig. 1: High-level Architecture of the proposed Method. To perform low-frequency compression enhancement and fully use the low-frequency information as prior for hyper network, we introduce the FSM and LFE Network to the compression model.

In conventional image and video compression methods, some extensive tools are proposed to utilize features of screen content [28] [32]. Intra Block Copy (IBC) involves the concept of motion estimation and compensation in intra-prediction because the replicate blocks often exist in the same screen content image. In addition, Palette Mode (PLT) is developed due to the limited number of colors that exist in Coding Units (CU), and Transform Skip (TS) depends on the fact that some types of screen content may not gain any benefit from the spatial transform.

Despite the relatively late start of learning-based image compression algorithms, they present great potential and the state-of-the-art (SOTA) has already surpassed conventional methods. Recently, the variational auto-encoder (VAE) method [17] has demonstrated superior rate-distortion performance compared to traditional lossy image compression algorithms, as evaluated by metrics such as Signal-to-Noise Ratio (PSNR) and Multi-Scale Structural Similarity Index Measure (MS-SSIM). This highlights its significant potential for practical

applications. The initial encoding involves utilizing both linear and nonlinear parametric analysis transformations to project the image into a latent space. Following quantization, the distribution of the latent variables is predicted using entropy estimation modules. Subsequently, the latent variables are compressed into a bit stream using either lossless Context-based Adaptive Binary Arithmetic Coding (CABAC) [23] or Range Coder [24]. Utilizing the VAE framework, enhancements such as hyper prior [4], auto-regressive priors [25], and Gaussian Mixture Model (GMM) [7] are incorporated to refine the precision of predicting the latent variable distribution by the entropy estimation modules, thereby achieving improved rate-distortion (RD) performance.

For decoding, the bit stream is decoded to recover the decompressed latent variables using either lossless CABAC or Range Coder. These latent variables are then mapped to reconstructed images through a combination of linear and nonlinear parametric synthesis transformations. To further enhance the quality of the reconstructed images, there is ongoing research into designing modules that function as post-processing components, such as artifact removal [1, 10, 12, 20, 22, 31, 34], super-resolution [11, 15], and deblocking [2, 18].

While existing methods for image compression have delivered excellent results, they tend to overlook the unique characteristics of screen content images. Thus, inspired by the idea of LC-FDNet [29] and the theoretical basis of SCC extensions of traditional coding methods, we propose an end-to-end leaning-based method for screen content image compression to provide a more efficient approach for compression screenshots and other forms of screen content images. The general network has a similar architecture to VAE. However, a FSM is involved, which decomposes the high and low-frequency area from the input image and sends low-frequency information to the Low-Frequency Enhancement branch. This design fully considers how to improve the compression rate of the low-frequency information, which takes up a large area of the screen in the practical use cases, and the outcome of the main encoder is used as a prior in the hyper-network for further processing. Our work's primary contribution is designing an effective network architecture that considers the unique characteristics of screen content images and improves RD performance by enhancing the low-frequency area.

2 Related Work

2.1 Learned Image Compression

In recent years, the CNN-based learned image compression models have flourished and have had outstanding breakthroughs [3, 4, 7–9, 14, 21, 25, 26, 30, 33]. Following the introduction of the first end-to-end optimized image compression model based on the VAE architecture [3], numerous subsequent studies have made modifications and sought to enhance the architecture. In [4], a hyper-prior was introduced to capture spatial dependencies within the latent representation. Building on this, [25] drew inspiration from the effectiveness of auto-regressive priors in probabilistic generative models. They extended the model proposed

by [4] by incorporating an auto-regressive module to further enhance its capabilities. Furthermore, Cheng et al. [7] introduced a network architecture that includes residual blocks and integrates a simplified attention module, along with using a Gaussian Mixture Model (GMM) instead of the traditional Single Gaussian Model (SGM). While the GMM enhances rate-distortion (RD) performance, it significantly increases encoding and decoding time consumption compared to the computationally less expensive SGM-based entropy model. To speed up the serial processing in autoregressive context models, [14] designed a checkerboard context model (CCM) from the spatial-wise perspective, which highly reduces the decoding time.Additionally, from a channel-wise auto-regressive perspective, the modified model employs a context model along the channel dimension rather than the serial-decoded spatial ones. This approach not only enhances the efficiency of both encoding and decoding processes but also improves the RD performance [26]. After that, [13] proposed a spatial-channel contextual adaptive model to combine the merits of both spatial-wise and channel-wise ideas. Furthermore, adding an informer module to exploit global information complementing the local information collected by the context module [16], using the latent of the low-frequency region as a strong prior for high-frequency region encoding [29], and introducing a Swin-Transformer-based attention (SWAtten) module with channel squeezing for optimizing channel-wise auto-regressive entropy models, as proposed by Liu et al. [19], represents another approach to enhance rate-distortion (RD) performance and reduce computation time in learned image compression models.

3 The Proposed Methods

In this paper, we present a Low-Frequency Enhancement (LFE) approach within an end-to-end coding framework. Our goal is to fully leverage the unique characteristics of screen content images and enhance the RD performance specifically for screen content image compression. Fig. 1 illustrates the high-level architecture of the proposed method, which contains the modules and layer parameters. The proposed LFE-SCCM implementation is on top of the Minnen2018 in [25], and Fig. 2 details the architecture of the proposed model.

3.1 Low-Frequency Enhancement

In the context of image compression, high-frequency and low-frequency information are two fundamental concepts that play a crucial role in capturing the essence of an image. On the one hand, low-frequency components represent the smooth and gradual changes in an image, such as large areas of uniform color or subtle variations that form the background and overall structure of the visual content.

On the other hand, high-frequency components contain finer details and rapid changes in pixel values, such as edges, texts, and sharp contrasts. These elements

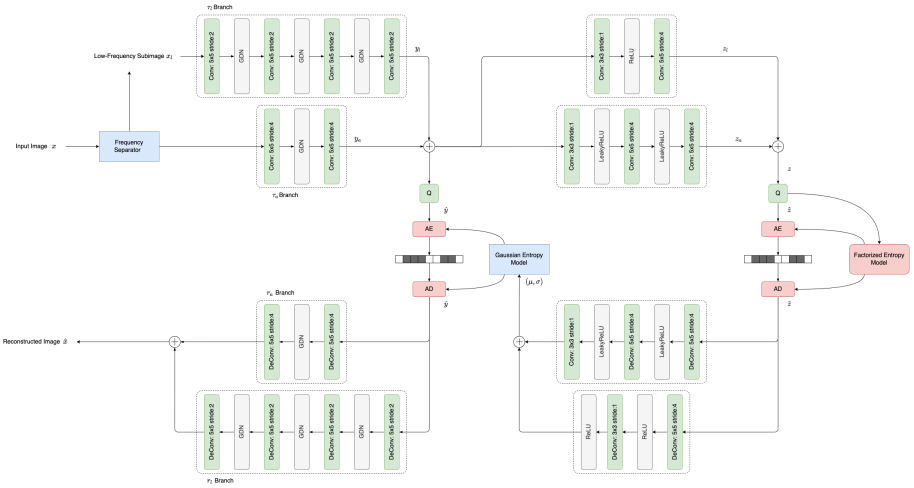


Fig. 2: Illustration of the proposed framework, where "Q" represents the quantization process, while "AE" and "AD" stand for the arithmetic entropy encoding and decoding processes, respectively.

are less predictable due to their spatially localized nature and the abrupt transitions between local pixel values. The presence of high-frequency information is essential for maintaining the visual acuity and fidelity of a screen content image, as it conveys the intricate details that define the content of the image.

In conventional compression methods, the usage of transformation and quantization to drop the high-frequency information is a common approach to reduce the bitrate. The coding algorithms use transformation functions to separate information by different frequencies and apply diverse quantization parameters according to the sensitive degree of the human vision system. While the loss of high-frequency information may lead to the deterioration of edges and details, low-frequency information contains the main content of an image or a frame.

However, the research of LC-FDNet [29] has shown that the relationship between high and low-frequency information is not mutually exclusive. In fact, the low-frequency components can serve as a foundation for the high-frequency details. By first encoding the low-frequency information, one can leverage its predictability and spatial consistency to assist the more complex high-frequency elements compression.

In the proposed method of screen content image compression, we prioritize low-frequency information with finer processing to ensure the reconstruction quality. Meanwhile, taking the concatenated outcome of the LFE branch and the main forward transform branch as a prior in the hyper codec will be helpful to the high-frequency reconstruction.

To obtain the low-frequency subimage, we have designed an FSM that is integrated into the network ahead of the main encoder. The FSM is responsible

for separating the low and high-frequency information. The FSM smooths the input image using Gaussian Blur to obtain the low-frequency component:

$$x_l = Gaussian(0.299 \times x_R + 0.587 \times x_G + 0.114 \times x_B) \tag{1}$$

where x_R, x_G and x_B are three color channels of the input image. $Gaussian$ is the Gaussian Blur operation and x_l represent the low-frequency subimages.

3.2 Network Architecture

In general, the end-to-end compression framework comprises an encoder and a decoder. Before an image is sent to the encoder, the FSM separates the low and high-frequency subimages and takes the low-frequency one, x_l, as another encoder input.

On the encoder side, there are two analytical branches involved for the input images x and x_l. These branches are the primary forward transform branch, denoted as τ_a, and the LFE branch, denoted as τ_l. Since the low-frequency components x_l are utilized as priors for the hyper network, a detailed processing approach is employed in τ_l. Specifically, this involves applying four convolutional layers with a stride of two sequentially. The primary forward transform branch τ_a employs a more streamlined approach, utilizing two convolutional layers with a stride of four. In other words, the origin image x is coarsely processed. The τ_a and τ_l branches can be formulated as follows:

$$y_a = \tau_a(x, \theta_a), \tag{2}$$

$$y_l = \tau_l(x_l, \theta_l), \tag{3}$$

where the latent codes are represented by y and y_l. The model parameters for the primary forward transform and the LFE branches are denoted as θ_a and θ_l. The total latent code y is formed by combining y_a and y_l, as described below:

$$y = y_a + y_l \tag{4}$$

the total latent is quantized, which is denoted as $\hat{y}$ and generated as follows,

$$\hat{y} = \mathcal{Q}(y) \tag{5}$$

the $\hat{y}$ is coded by the AE.

As the proposed method is based on the Minnen2018 model in [25], the means and scales are predicted by the hyperprior model. In our method, the hyperencoder and decoder are extended with low-frequency enhancement branches. The hyperlatent in terms of z is composed of outputs of two branches, z_a and z_l, as follows,

$$z = z_a + z_l \tag{6}$$

After that, quantization and the factorized entropy model [4] cooperate to compress the z.

Building on the work of Minnen et al. [25], each latent variable $\hat{y}_i$ is modeled as a Gaussian distribution with mean μ_i and standard deviation σ_i, convolved with a unit uniform distribution:

$$p_{\hat{y}|\hat{z},\theta} = \prod_{i=1} \left((\mathcal{N}(\mu_i, \sigma_i^2) * \mathcal{U}(-0.5, 0.5)) \right) (\hat{y}_i) \tag{7}$$

where μ and σ are the predicted parameters of the entropy model, $\hat{z}$ represents the quantized hyper-latents, and θ denotes the parameters of the entropy model.

To optimize end-to-end learning compression models, the training process incorporates rate and distortion optimization with Lagrangian parameters. Distortion is quantified as the difference between the reconstructed image $\hat{x}$ and the original input image x, specifically using Mean Square Error (MSE) The rate term includes the bits needed to encode $\hat{y}$ and $\hat{z}$. Consequently, the loss function is defined as follows:

$$\mathcal{L} = \mathcal{R}(\hat{y}) + \mathcal{R}(\hat{z}) + \lambda \cdot \mathcal{D}(x, \hat{x}) \tag{8}$$

where λ denotes the Lagrange parameter, which controls the trade-off between rate and distortion.

4 Experiment

In this section, we present the experiments and results of the proposed LFE-SCCM model, along with a comparison of its performance against existing image compression models. To analyze the effectiveness of the proposed model architecture on the screen content image compression, both models taken as the benchmark are retrained on the LSCD dataset [6].

4.1 Experimental Setup

Datasets. We train all three models on the images collected in the LSCD dataset [6], which is a screen content image dataset considering practical computer use cases. As it is a dataset for 1080P screen content video compression, some of the frames are collected as training and testing images. In our setting, we use 3,096 images from the LSCD dataset for training. We evaluate the rate–distortion performance on the SCCM dataset we collected, which covers diverse content in practical usage, such as video watching, web surfing, document reading, and MS Office processing, with a fixed resolution of 1920 × 1080. In total, we use 128 images from the SCCM dataset as test images.

Training. The implementation of our model is based on PyTorch [27] and an open-source compression project, CompressAI [5]. all three models are trained for 3000 epochs with a batch size of 16. To streamline the training process, images in both the training and validation sets are cropped into 256 × 256 patches. During

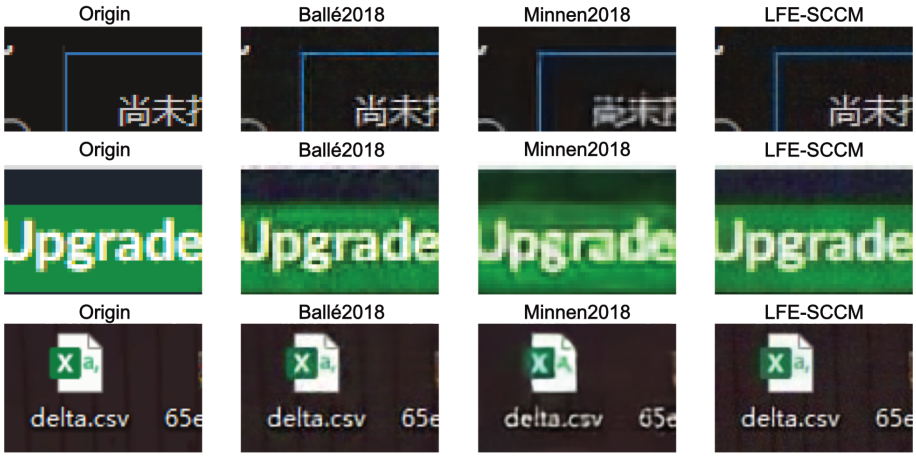


Fig. 3: The visual comparison.

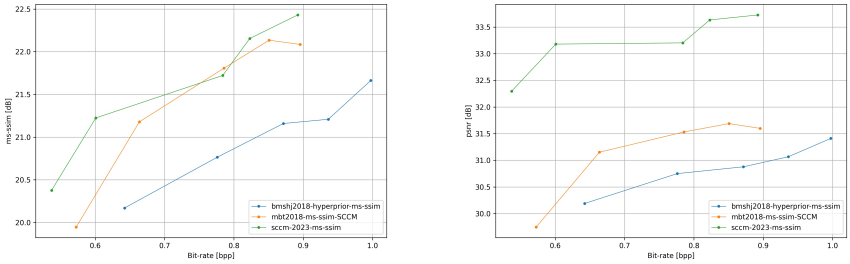
training, the loss function utilizes mean square error (MSE), with λ values set to 0.02, 0.03, 0.04, 0.05, 0.06 to cover a broad spectrum of bit-rates for compression. In this setup, five models are trained, each representing different quality levels. We use the Adam optimizer throughout training. When executed on a GeForce RTX 3090 Ti, the training process typically lasts around 12 hours.

Evaluation. We assess the performance of our LFE-SCCM by computing the average RD performance metrics, including PSNR and MS-SSIM, on the SCCM dataset. This dataset encompasses four practical use cases of computers in daily life. We conduct comparisons with prominent general learned compression methods, including the Ballé2018 model [4] and the Minnen2018 model [25]. To compare and evaluate their performance on screen content image compression, both of them are retrained on the same dataset as the proposed method with 3000 epochs.

4.2 Results

Fig. 3 displays the visual outcomes of various models. When compared to the other models considered, our proposed LFE-SCCM demonstrates superior performance with screen content images. Furthermore, as depicted in Fig. 4, the retrained Minnen2018 model [25] exhibited better performance than the retrained Balle2018 model [4]. The outcomes align with the trends reported in the original publication by Minnen et al. [25], where the model was trained using a dataset comprising natural images.

To demonstrate the superior performance of our model, we used the commonly used metrics MS-SSIM and PSNR to evaluate the reconstruction performance. As shown in Fig. 4a, our LFE-SCCM model performed better than the retrained Minnen2018 model [25] in the low-bpp and high-bpp regions. However, in the middle-bpp region, the reconstructed image quality of our model



(a) (b)

Fig. 4: Rate-Distortion (RD) performance comparison of LFE-SCCM against the retrained Ballé2018 [4] and Minnen2018 [25] models. (a) RD-curves for MS-SSIM. (b) RD-curves for PSNR.

was slightly inferior to the retrained Minnen2018 model [25] when evaluated by MS-SSIM. In Fig. 4b, it is apparent that our LFE-SCCM model outperformed both of the other retrained models on PSNR for all bpp values.

5 Ablation Study

Table 1 presents the findings from an ablation study designed to assess the impact of Low-Frequency Enhancement on rate-distortion performance. The baseline model, which shares the same architecture, differs in that it does not utilize the low-frequency as the input for the t_l branch. This model disregards the significance of low-frequency information when compressing the high-frequency region by using the original image as input for both branches, thereby equating the importance of both frequency areas. This validates the effectiveness of the low-frequency enhancement branch in improving compression efficiency.

Method	λ	**bpp**	**PSNR**
Baseline	0.02	0.690	32.26
LFE-SCCM	0.02	0.533	32.29
Baseline	0.04	1.039	32.95
LFE-SCCM	0.04	0.600	33.20

Table 1: Comparison of Methods at Different λ Values

The results, obtained under the same λ conditions, clearly demonstrate that assigning greater emphasis to low-frequency information and employing it as a prior for high-frequency compression leads to a dual benefit: an increase in PSNR and a reduction in bpp. This outcome underscores the pivotal role of

low-frequency information, even in images with a greater prevalence of high-frequency content, such as screen content images. The persistent relevance of low-frequency data highlights its indispensable contribution to overall compression efficiency and quality preservation.

6 Conclusion

In this paper, we present an innovative learning-based method called LFE-SCCM for compressing screen content images. By combining principles of traditional compression and screen content characteristics, we introduce a Low-Frequency Enhancement branch that extracts low-frequency information as a prior for the hyperprior network, while the primary branch processes the full image. Extensive experiments show that low-frequency information plays a significant role, and our method achieves promising compression improvements.

However, performance on game-type screen content is less competitive, as such images contain richer textures similar to natural images.

For future work, we plan to explore adaptive frequency-aware modules for game-like content and investigate a Stable-Diffusion–like strategy to progressively generate high-frequency details from low-frequency representations, aiming to further enhance reconstruction quality.

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